ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Formation of new substance indicates chemical change.
2	**(a)** Both A and R are true and R is the correct explanation of A.	H^+ is needed for acidic behaviour.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene prevents contact with air/water.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Weak forces need little energy.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Low oxygen requires fast breathing.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Spinal cord reflex centre.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Copying errors produce small variations, which explains the statement.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Fusion with X or Y sperm decides the sex.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The properties of the convex mirror explain its use.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why the image is inverted.
11	**(a)** Both A and R are true and R is the correct explanation of A.	High resistance draws almost no current.
12	**(a)** Both A and R are true and R is the correct explanation of A.	At a crossing there would be two directions of the field.
13	**(a)** Both A and R are true and R is the correct explanation of A.	The energy left is too little to support further levels.
14	**(d)** 44 g	Mass is conserved: 12 + 32 = 44 g.
15	**(d)** 7	Complete neutralisation to NaCl.
16	**(a)** Y	Reacting with cold water indicates more reactivity.
17	**(c)** $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$	Atoms of C, H and O balance.
18	**(d)** 6	$C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O$.
19	**(d)** diabetes	Insulin.
20	**(b)** variation, and those from the cutting are identical to the parent	Sexual gives variation; vegetative does not.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Mass of CO_2 = 12 + 32 = 44 g (1); Law of conservation of mass (1).	1+1
22	CO_2 (1); $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$, lime water turns milky (1).	1+1
23	Oxide is amphoteric (1); Zinc or aluminium (1).	1+1
24	$2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$ (1); Products: carbon dioxide and water (1).	1+1
25	Volume per minute = 72 x 70 (1); = 5040 mL (5.04 L) (1).	1+1
26	Motor neuron or effector (1); Sensory path is intact (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks

27	Equation already balanced: $PbO + C \longrightarrow Pb + CO$ (1); PbO is the oxidising agent (1); C is the reducing agent (1).	1+1+1
28	pH 10 and 13 turn it pink (1); pH 1 and 4 neutralise a base (1); Acidity order: pH 1 > 4 > 7 > 10 > 13 (1).	1+1+1
29	ZnO is the oxidising agent and C is the reducing agent (1); Redox, reduction of metal oxide with carbon (1); Zinc is moderately reactive, so carbon can reduce its oxide (1).	1+1+1
30	$CH_3-CH=CH_2$ (1); Propene is unsaturated (a double bond); propane is saturated (1); Propene undergoes addition reactions; propane undergoes substitution (1).	1+1+1
31	Light is needed for photosynthesis (1); Starch is formed in light only (1); Covered region shows no starch (1).	1+1+1
	OR: Tip makes auxin (1); Auxin goes to the shaded side (1); No tip: no auxin, no bending (1).	1+1+1
32	Divisions in 2 hours = 8 (1); Number = 2^8 = 256 (1); Binary fission (asexual) (1).	1+1+1
	OR: Cross with green wrinkled (yyrr) (1); All yellow round offspring means YYRR (1); Other phenotypes in offspring mean the plant is heterozygous (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	A: no rust (1); B: no rust (1); C: rust forms (1); Conclusion: air (oxygen) and water are both needed (1); Prevention: galvanisation/painting (1).	1+1+1+1+1
34	HCl pH lower (1); HCl conducts better (1); Mg reacts faster with HCl (1); HCl is a strong acid ionising completely (1); CH_3COOH is weak, ionising partially (1).	1+1+1+1+1
35	Order: A > B > C > D (1); A: electrolysis of molten salt (very reactive) (1); D: occurs free or by heating its ore (1); Example set: Mg, Zn, Cu, Ag (1); Reason: a more reactive metal displaces a less reactive one (1).	1+1+1+1+1
36	Dehydration: conc. H_2SO_4, 443 K (1); Oxidation: alkaline $KMnO_4$ (1); $KMnO_4$ is oxidising agent (1); Esterification: conc. H_2SO_4 catalyst (1); Equations of each step (1).	1+1+1+1+1
	OR: Aim: to show CO_2 is needed for photosynthesis (1); KOH absorbs CO_2 (1); Leaf with KOH: no starch (1); Leaf with water: starch present (1); Control: leaf in water flask (1).	1+1+1+1+1
37	Aim: role of tip in phototropism (1); Take three seedlings: normal, tip cut, tip covered with foil (1); Expose to light from one side (1); Observation: only normal seedling bends (1); Conclusion: tip senses light and makes auxin (1).	1+1+1+1+1
	OR: Asexual offspring are nearly identical clones (1); DNA copying errors give small variations only (1); Sexual reproduction mixes DNA of two parents (1); Variations help some survive changes in the environment (1); Example: cross-pollination in plants gives varied seeds (1).	1+1+1+1+1
38	F1: TtRr, tall round (1); F2 ratio 9 : 3 : 3 : 1 (1); Tall round 900, tall wrinkled 300 (1); Dwarf round 300, dwarf wrinkled 100 (1); Dwarf round genotypes ttRR and ttRr (1).	1+1+1+1+1
	OR: Concave: 1/v = -1/20 + 1/30 = -1/60, v = -60 cm (1); m = -2, size 8 cm, real, inverted (1); Convex: 1/v = 1/20 + 1/30 = 1/12, v = +12 cm (1); m = +0.4, size 1.6 cm (1); Convex: virtual, erect, diminished (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 50	1/4 of 200 = 50 (Rryy).
2	**(b)** 11.25 cm	1/f = -1/45 - 1/15 = -4/45, so f = -11.25 cm.
3	**(d)** -25 cm	f = -(far point) for objects at infinity.
4	**(c)** 6 ohm	R2 = rho 2L / 4A = R / 2.
5	**(b)** 90 degrees	Perpendicular is the maximum.
6	**(a)** 5000 J	5 x 10 x 10 x 10.
7	**(b)** copper	CuO is reduced to Cu by hydrogen.
8	**(a)** plaster of Paris	It sets into gypsum.
9	**(d)** CuO and CO_2	Calcination: $CuCO_3 \longrightarrow CuO + CO_2$.
10	**(d)** ethene	C_2H_4 = 28.
11	**(a)** uses carbon dioxide and releases oxygen	Photosynthesis replenishes oxygen.
12	**(c)** adrenaline	Fight or flight.
13	**(d)** undifferentiated cells to produce missing parts	Specialised dividing cells are essential.
14	**(d)** 1 : 1 : 1 : 1	Four gamete types x one.
15	**(b)** convex, f = 12 cm	Object at 2F.
16	**(b)** smaller wavelengths are scattered more by small particles	Rayleigh type scattering.
17	**(a)** V against I	Ohm's law.
18	**(c)** 100 Hz	Proportional to speed.
19	**(a)** UV radiation on oxygen	O2 splits and recombines.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Energy is needed to break bonds.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Regeneration (1); It involves growth of a whole body from a body part, not an outgrowth (1).	1+1
22	TT fraction = 1/4 (1); 60 plants (1).	1+1
23	m = -1/3 = -v/u, v = -15, so u = -45 cm (1); 1/f = -1/15 - 1/45 = -4/45, f = -11.25 cm (1).	1+1
24	Red is scattered the least (1); The scattered light is dominated by shorter wavelengths (1).	1+1
25	I = 4.8×10^{-3} A (1); n = 4.8×10^{-3} / 1.6×10^{-19} = 3×10^{16} (1).	1+1
26	Force is perpendicular to the velocity (1); No work is done, so the speed is constant (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	First case: u = -x, v = -3x, so 1/f = -1/(3x) - 1/x = -4/(3x), |f| = 3x/4 (1); Second case: same size means the object is at C, so x + 10 = 2|f| = 3x/2 (1); x = 20 cm, so f = -15 cm (1).	1+1+1
28	1/f = -1/30 + 1/25 (1); 1/f = 1/150, f = 150 cm (1); P = +0.67 D (1).	1+1+1
29	Atoms vibrate more, increasing collisions with electrons (1); Resistance increases (1); Current decreases (I = V / R) (1).	1+1+1
30	Force arises from the component of the field perpendicular to the wire (1); It is maximum when the wire is perpendicular (1); It is zero when the wire is parallel (1).	1+1+1
31	Small fish: 0.2 ppm (1); Big fish: 2 ppm (1); Kingfisher: 20 ppm (1).	1+1+1
	OR: Basic copper carbonate (1); Copper reacts with moisture, CO_2 and oxygen (1); Prevent by lacquer/paint or keeping dry (1).	1+1+1

32	X is $CaCO_3$, Y is CaO (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1).	1+1+1
	OR: Salt water increases electrical conductivity (1); Faster electrochemical corrosion (1); Rust formation speeds up (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(a) f = -90 cm, P = -1.11 D (1); (b) Concave lens (1); (c) f = -100 cm; image must be at 90 cm (1); 1/u = 1/v - 1/f = -1/90 + 1/100 = -1/900 (1); New far point is 900 cm (9 m), so correction is incomplete (1).	1+1+1+1+1
34	(a) Parallel part = 48 / 16 = 3 ohm; total = 9 ohm (1); (b) I = 12 / 9 = 1.33 A (1); (c) Voltage across parallel = 1.33 x 3 = 4 V (1); Currents: 4/12 = 0.33 A and 4/4 = 1 A (1); (d) P = 12 x 1.33 = 16 W (1).	1+1+1+1+1
35	Fleming's left-hand rule (1); Forefinger north, middle finger east (1); Thumb points upward (1); (a) Force becomes downward (1); (b) Both reversed: force upward as before (1).	1+1+1+1+1
36	Zooplankton: 200,000 J (1); Small fish: 20,000 J (1); Large fish: 2000 J; birds: 200 J (1); Energy lost at the first transfer = 1,800,000 J (1); The first transfer has the largest loss (1).	1+1+1+1+1
	OR: Green to white (water lost), then reddish brown residue (1); $2\,FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$ (1); Gases: SO_2 and SO_3 (1); Smell of burning sulphur (1); Thermal decomposition (1).	1+1+1+1+1
37	Fixed H2O molecules per formula unit (1); $CuSO_4 \cdot 5\,H_2O$: blue to white (1); $FeSO_4 \cdot 7\,H_2O$: green to white then brown (1); $Na_2CO_3 \cdot 10\,H_2O$ loses water to give a white powder (1); Test: add water to the white residue - blue colour for $CuSO_4$ (1).	1+1+1+1+1
	OR: (a) Reacts with air and water (1); (b) Malleable, forms Al_2O_3 layer (1); (c) Pure gold is soft (1); (d) Like ions repel when layers shift (1); Each reason relates to property (1).	1+1+1+1+1
38	Butane and isobutane (1); But-1-ene and but-2-ene (1); Isomers: same formula, different structure (1); Homologues: differ by CH_2 (1); Examples (1).	1+1+1+1+1
	OR: Aerobic: mitochondria, CO_2 + H_2O (1); Yeast: ethanol + CO_2 (1); Muscle: lactic acid (1); Only 2 ATP in anaerobic (1); Complete oxidation yields about 30 ATP (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Heat could cause splashing of acid.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Oxide layer protects the metal.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Incomplete combustion gives soot.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Valves prevent backflow.
5	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true but not an explanation of neuron direction itself.
6	**(d)** A is false but R is true.	A is false: most organisms cannot reproduce by regeneration; R is true.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Germ-cell DNA is unchanged, so acquired traits are not passed on.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Refracted rays only appear to meet when extended backwards.
9	**(c)** A is true but R is false.	A is true; R is false: the image forms in front of the retina.
10	**(c)** A is true but R is false.	A is true; R is false: copper has low resistivity.
11	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the deflection is evidence of the field, not its explanation.
12	**(a)** Both A and R are true and R is the correct explanation of A.	Decomposition recycles nutrients.
13	**(b)** 16 g	40 - 24 = 16 g.
14	**(a)** 25 mL	Ratio 1 : 2.
15	**(d)** S	Each displaces the next, so S is lowest.
16	**(c)** 6	Given in the equation.
17	**(d)** 1	$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$.
18	**(a)** reflex action is faster than brain sensation	Information reaches the brain later.
19	**(c)** spores need moisture to germinate	Water is needed to activate spores.
20	**(a)** 1/4	Each birth is independent.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Grass = 50 x 100 = 5000 J (1); Snake = 5 J (1).	1+1
22	Gases NO_2 and O_2 escape (1); Only solid PbO remains; the total mass is still conserved (1).	1+1
23	Plaster of Paris 145 g (1); Water lost 27 g (1).	1+1
24	Reduction of oxide is easier by carbon (1); So sulphides/carbonates are first converted to oxides (1).	1+1
25	$3n + 1 = 22$, n = 7 (1); C_7H_{16} (1).	1+1
26	Bile neutralises acid from the stomach (1); Pancreatic enzymes work in alkaline medium (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	n-butane: 3 C-C, 10 C-H (1); isobutane: 3 C-C, 10 C-H (1); Same number of bonds; isomers (1).	1+1+1
28	Heart works continuously (1); Needs oxygen and glucose (1); Blockage leads to heart attack (1).	1+1+1
29	Motor nerve carries signal to muscle (1); Effector is the muscle (1); Sensory part not needed (1).	1+1+1

30	Plants have meristematic cells for new growth (1); Humans have complex organ systems (1); Human body parts cannot make a whole body (1).	1+1+1
31	Sperm 23 (22 autosomes) (1); Egg 23 (22 autosomes); zygote 46 (44 autosomes) (1); Liver cell 46 (44 autosomes) (1).	1+1+1
	OR: m = -5, v = -5u (1); 1/12 = 1/(5|u|) - ... with u negative: 1/12 = 1/(5|u|) + 1/|u| = 6/(5|u|) (1); |u| = 14.4 cm, v = 72 cm (1).	1+1+1
32	Light reaches us after passing through a thick layer of air (1); Light is scattered many times by particles of all sizes (1); Colours mix and appear whitish (1).	1+1+1
	OR: P = 300 W (1); I = 300 / 220 = 1.36 A (1); R = 220 / 1.36 = 161.3 ohm (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Root bends down (1); Shoot bends up (1); Auxin collects on lower side (1); High auxin promotes shoot growth, inhibits root (1); Gravity-induced response (1).	1+1+1+1+1
34	Egg is released from the ovary into the oviduct (1); Sperm fuses with the egg in the oviduct (1); Zygote divides and moves towards the uterus (1); Embryo is implanted in the thick lining (1); Placenta forms from embryonic and uterine tissue (1).	1+1+1+1+1
35	First two plants: dwarf offspring means each has t (1); 1/4 dwarf shows both are Tt (1); Third plant crossed with tt gives all tall (1); So it must be TT (1); If it were Tt, half would be dwarf (1).	1+1+1+1+1
36	Now 1/v = -1/20 + 1/30 = -1/60, v = -60 cm (1); For image at infinity the object must be at F, 20 cm (1); Move 10 cm towards the mirror (1); Reflected beam is parallel (1); Used in torches and headlights (1).	1+1+1+1+1
	OR: u = infinity, v = -2 m (1); f = -2 m (1); P = -0.5 D (1); Image at the far point, 2 m (1); It is virtual and erect for the lens (1).	1+1+1+1+1
37	20 ohm: P = 14400 / 20 = 720 W (1); 30 ohm: P = 480 W (1); Series 50 ohm: 288 W (1); Parallel 12 ohm: 1200 W (1); Parallel gives the highest power (1).	1+1+1+1+1
	OR: Both have similar field patterns (1); Lines go from N to S outside (1); The solenoid's field can be switched on or off (1); A soft iron core makes it a strong electromagnet (1); Scrap is lifted while the current flows and dropped when switched off (1).	1+1+1+1+1
38	Pesticide kills insects and remains in their bodies (1); Birds that eat insects accumulate it (1); Biomagnification poisons the birds (1); Alternative: biological control with natural predators (1); Alternative: crop rotation or organic farming (1).	1+1+1+1+1
	OR: Zinc: bubbles of hydrogen (1); $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Copper: no reaction (1); Zinc is above hydrogen in the reactivity series (1); Copper is below hydrogen so cannot displace it (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.